

07a — Deleted / replaced content (ghost text)

What it detects: on a scan that was **converted to editable text** (Acrobat "Edit scanned document", iLovePDF and the like), text that was **deleted** or **shortened** after the conversion.

Code: `services/forensics/ghost_content.py` (`analyze_ghost_content` , `exclude_signature_regions`), run by `tasks/tampering_checks_task.py` (`run_tampering_checks` , same task as ELA / copy-move since it reads the same bytes). One `document_checks` row: `ghost_content` .

Why it works

A converter erases the text pixels from the scan and redraws every word as live text on top. The erasure is never perfect: a faint **ghost** of each word stays in the background image.

What the background shows	Meaning
ghost with live text on top	normal — the converted word
ghost with no live text on top	content that was there at conversion and was deleted since
ghost much wider than the live text on top	the line was shortened or rewritten

When it runs

Only on pages built that way (`page_structure.PageStructure.vector_text_over_image` , from PyMuPDF's drawing order): one image covering $\geq 90\%$ of the page (it may overhang the page edges, as converters place it — e.g. `-10, -7, 605, 849` on a 595×842 page) drawn **before** the text, and ≥ 15 visible text runs (render mode $\neq 3$) spanning $\geq 50\%$ of the page height — **whatever the fonts are called** (a converter's `-NNNN` subsets or ordinary `ABCDEF+` subsets).

A plain scan (no text layer), a searchable scan (OCR text invisible, render mode 3) and a born-digital page (no page-sized image) → result `not_applicable` , with the reason in `data.reason` ("no text layer (a plain scan)", ...) — shown on the case page and in the report's coverage table ("Not applicable: no text layer (a plain scan)").

Documents processed before this check existed: `python -m scripts.backfill_ghost_content` (all of them, or `--case CASE-...`) runs it on every PDF without a result, and re-scores a case that flags. A converted page whose scan is placed rotated or skewed is also reported `not_applicable` (not handled).

Algorithm (background image at its native resolution, 8-bit gray)

- 1. Residue.** Morphological **black-hat** with a kernel of $\frac{1}{2}$ the median text height: how much darker each pixel is than the paper around it, for marks thinner than a stroke. Broad shading and uneven illumination are part of the "paper".
- 2. Not searched:** where the paper itself is not white — shaded cells and bands (often a scanner's halftone dots), photos, logos left in the scan: median tone over ~ 2 text heights more than 12 levels darker than the page's white. A ghost cannot be told from that texture.
- 3. Ghost band.** Residue between the page's noise floor (median over empty 64-px tiles of their 99th percentile, at least 4) and **70** (stronger is ink still in the image — rules, logos, stamps, signatures — removed with a 2-px halo). Ruled lines are removed (bridged across short breaks, then long thin runs; a band as tall as letters is text, not a rule), as are specks < 6 px and components much flatter than a glyph.
- 4. Lines.** The ghost mask is smeared horizontally ($0.6 \times$ text height) into lines; kept when $0.5\text{--}2.5 \times$ the median text height tall, ≥ 25 pt wide, aspect ≥ 1.8 and 3–60 % ink.

5. **What may sit on a ghost:** visible text spans (+3 pt; lines skewed by up to ~6° count), other images (+2 pt — converters lift logos, stamps and signatures into their own images), filled vector shapes, and a 4 % page margin.
6. **Classify each ghost line** (after dropping a line with more ink still in the image than ghost — the faint edge of a signature or of print the converter left as image — and a line $\geq 1.5 \times$ stronger than the page's own erased text, i.e. a smudge):
 - **replaced line** (`ghost_replaced_line`): its *uncovered* ghost runs on past the live text on its row (or before it) by $> 30 \%$ of the end word and > 15 pt, into empty space (not up to the next word), and that uncovered part itself looks like text (≥ 25 pt, $\geq 50 \%$ of its columns marked, ≥ 4 glyph-height marks). Text the converter could not map to characters is skipped (its width means nothing).
 - **orphan line:** $< 20 \%$ covered, no live word on its row within 4 text heights, and text-like (≥ 3 glyph-height marks). Orphan lines less than 1.5 line heights apart, side by side within 3 line heights (a label column and its value column), merge into a **block**; a block needs one line ≥ 40 pt.
7. **Show-through** (when OCR is available): each orphan block's enhanced crop is read as is and mirrored (Azure Document Intelligence Read). If the mirrored reading is real text (≥ 4 characters, confidence ≥ 0.5) and clearly better ($\times 1.5$), it is the back of the page showing through: `ghost_show_through`, info, not scored. The as-is reading is kept as the reviewer's hint — best effort; a heavily compressed scan often gives none.
8. **Vision-model hint** (when an LLM with vision is configured): for each deleted block, the trace and a tiny layout thumbnail of the page (too small to read — the model must not take the surrounding text for the block's) go to the vision model **three times** (`guess_erased_content`, temperature 0.4). Only what at least two answers agree on is kept (`data.vision_hint`: `kinds`, `heading`, `legible_words`, `answers_agree`); confidence is "medium" only when the answers agree on the kind and read something consistently, else "low". On CASE-39CB18BF the answers disagree (bank details / a note about fees), so it shows as a low-confidence guess. Never scored, never evidence.
9. **Signatures and stamps:** a finding lying $\geq 30 \%$ inside a detected signature or stamp region is set aside (`ghost_near_signature`, info). Applied when the ghost check runs if detection has finished, else when detection finishes (`refilter_ghost_content`, called from `signature_detection_task`).

Findings

Finding	Severity	Meaning
<code>ghost_content_scope</code>	info	Pages searched (noise floor and erased-text level per page), or why not applicable
<code>ghost_deleted_block</code>	high	A block of faint text with no live text on top: <code>data.lines</code> , <code>data.box_pt</code> , <code>data.hint</code>
<code>ghost_replaced_line</code>	medium	A line's trace runs on past its live text: <code>data.text</code> , <code>data.overhang_pt</code>
<code>ghost_show_through</code>	info	An orphan block that reads better mirrored — not scored
<code>ghost_near_signature</code>	info	Set aside: inside a detected signature or stamp — not scored

Every located finding carries a normalized `bounding_box` and `data.crop_png_base64` — the scan in that box, contrast-stretched (darkest 0.5 % black, the paper's median tone white) so the trace shows. The case page draws deleted / shortened content as a **solid teal** box and shows the crop under the finding; the report draws the same box (Section 9) and adds a page of crops after the annotated page.

`result` is `flag` when any finding is medium or high, `pass` otherwise, `not_applicable` as above. In the report's coverage table a check that was not applicable to any document reads "Not applicable".

Risk rules fed

Rule	Finding	Weight	Severity
<code>content.deleted_ghost_block</code>	<code>ghost_deleted_block</code> (high)	25	high
<code>content.replaced_ghost_line</code>	<code>ghost_replaced_line</code> (medium, high)	10	medium

Each fires at most once per document (the reason lists every block / line).

Reference results

Case	Result
CASE-39CB18BF (Ajyal, <code>6f277532...</code>)	flag: a 6-line block deleted under the fee table (x 30–270, y 388–480 pt — the bank-transfer details); the "Registration Fees" row's trace runs ~90 pt past its text
CASE-1CEDFDBF / CASE-F6F5FE76 (British Orchard, <code>895a259e...</code>)	pass — ghosts under the converted text only; the shaded cells, the footer the converter garbled and the stamp give nothing
CASE-B70765EC (ILM Academy, <code>14926b7f...</code>)	pass — no false orphans from the shaded rows
CASE-DE627FA2, CASE-DB43653A (MFP scans), CASE-4B297080 (Word)	not applicable

Known limitations

- **The student row of CASE-39CB18BF is not reported as shortened.** Its trace runs only ~6 pt past "PRE-KG-NA" (the words underneath are spaced differently from the live words, but are about as long); converter re-typesetting routinely differs by that much, so it cannot be told apart. The retyped name, ID and class are caught instead by the font check: they sit in a second embedded copy of the bold font (`font_subset_split` , 06a).
- Shaded areas are not searched, so content deleted from a shaded cell is missed.
- The hint OCR rarely reads anything on a heavily compressed scan; the crop is the evidence.
- A scan placed rotated / skewed (as an image transform) is not handled.